Final Project Design Phase

Computer Science - Week 11

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Having good, well documented code. Straightforward naming conventions and keeping things together in a meaningful way.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the coding to be the easiest, and the scoping (and Drafter implementation) to be the hardest. I think scoping may be hard for me since I tend to be overly ambitious, and Drafter implementation I think I do well at but is generally a bit unconventional and niche, so I expect to have to put a bit of effort into working it out.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Probably implementing the Drafter design, and potentially sourcing my information

depending on my scope/ ambitions.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Snowball effect– I tend to have new ideas on top of new ideas that gradually increase the scale of my projects, and at points it tends to get away from me.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Home Page

Search Bar

General description….

Contribute

…

High-traffic page

High-traffic page

High-traffic page

High-traffic page

References/ creator info

Search results

Search Bar

Example Page Result
Example page description … example description continued

Corresponding icon

Example Page Result
Example page description …

Corresponding icon

Example Page Result
Example page description …

Corresponding icon

Specific Page

Search Bar

Specified item of research

Description of item with some connecting mapping inline links for relevant words

Corresponding icon

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	It will serve as a location to amass and connect information under a given topic (namely, a video game, though the general premise is versatile) in an intuitive and readily available manner.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	I think my State dataclass could have several fields, including search_query, page, selected_category, message, articles, and current_article_title, just to name a few. These would be updated in the program– I believe storing them in state would be useful because they are updated and referenced frequently and are important.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	I think WikiArticle could be a good dataclass, since each has a lot of information I want to keep organized and use the template of for others. title:str description:str category:str related_terms:list[str] url:str I think these are self explanatory, and would be sue to keep information organized.
Routes (minimum 7)	not sure exactly what I'll decide on, but here are

The names of your route functions	some I might use: show_home search_wiki show_search show_article show_category show_contribution submit_contribution I'd use a parameter of state when calling the route functions, which are primarily used to change pages or go to different parts of the website.
Pages (minimum 5) A brief description of the pages in your site	One home page, an adaptive search results page, and adaptive item pages. While I'm not exactly sure how I want to make these pages yet as I haven't begun programming, I believe using a similar format for item result pages is good programming technique and is favored by a user. However, I believe it may be best for me to save these separate pages as their own individual pages, so in reality I would have several item pages. Additionally, I would like to have the feature to create a new page built into the website UI (not requiring programming).
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	The user can click several links (both as dedicated standalone buttons and as interactable links in text) to interact with the website. Additionally, they can use the search bar to type on their keyboard a search query. Also, if it counts (although it's probably built-in), the user will do general website-related things, such as using the mouse/ touchpad/ touchscreen to scroll up and down in the website for pages that have more vertical height than fits on the screen.

If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	I expect to have a plethora of conditional statements throughout the program. A few instances of these could manifest as being things like only taking user input for the searchbar after a user clicks on the searchbar with their cursor, handling matching names upon edits of the in-website UI (ie– a user updates Mana to Mana Meter, should change the name in all occurrences of its name), knowing when to change pages and what page to change to, and helping the user create a new page to define a new item. These if statements help structure the adaptive logic, allowing it to be interacted with by the user.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Iterating over a list of pages for objects and finding one in which the string given by the user in the searchbar best matches with the name of the item. This is essential to the search feature and directly integrates the iterative functionality of a loop.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use a LLM for this assignment!

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.